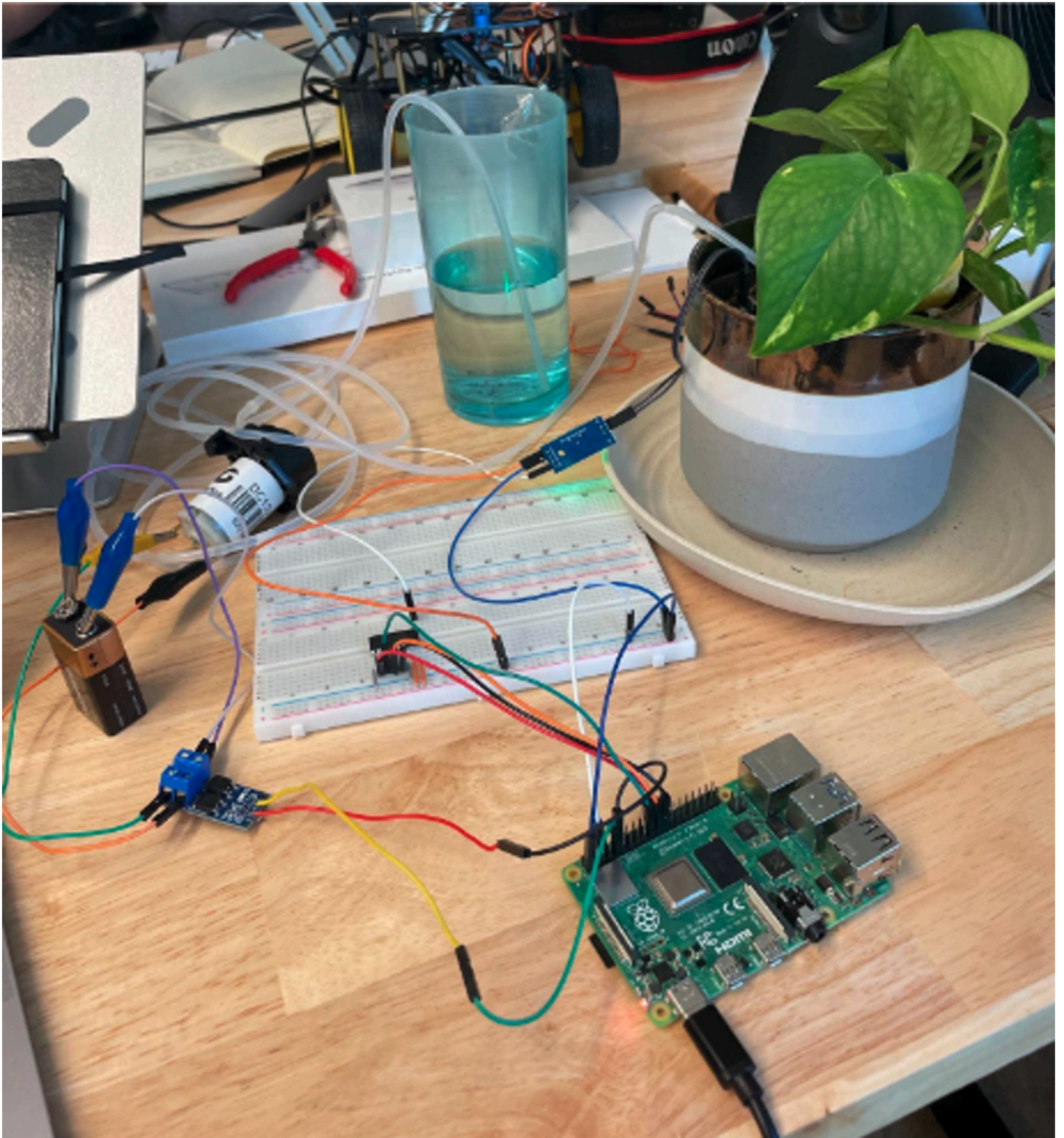


Plant Watering Robot with a Raspberry Pi



Building a useful robot doesn't have to require complicated code or expensive equipment. With a Raspberry Pi and some cheap, basic hardware, you can keep your plants healthy and happy from anywhere in the world!

Follow this tutorial to learn how to set up an automatic plant watering system:

- 1 [Complete the physical assembly and wiring.](#)
- 2 [Create and connect to the robot, and configure your robot's components.](#)
- 3 [Configure the ADC as a module from the registry.](#)
- 4 [Write code utilizing the Viam Python SDK to control the plant watering robot.](#)

Beginner-Friendly: Build an Automatic Plant Watering Robot



You can also follow a simplified version of this tutorial in this video: it eliminates the need for the ADC and the breadboard, instead using the digital pin of the moisture sensor to get “high” and “low” readings and to turn a relay on and off. You can start with this simple version and then add the ADC to make your machine more accurate!

The tutorial uses the following hardware, but you can adjust it as needed:

- A Raspberry Pi 3B or 4B with SD card and [5V USB power supply](#)
- A [resistive soil moisture sensor](#)
- A [5V mini water pump](#) with clear vinyl tube
- A [MCP3008 ADC](#)
- A [one channel relay switch](#)
- A [breadboard](#)
- Assorted [breadboard jumper wires](#), including wires with [alligator clips](#)
- A planter box or flower pot
- A water container
- A screwdriver

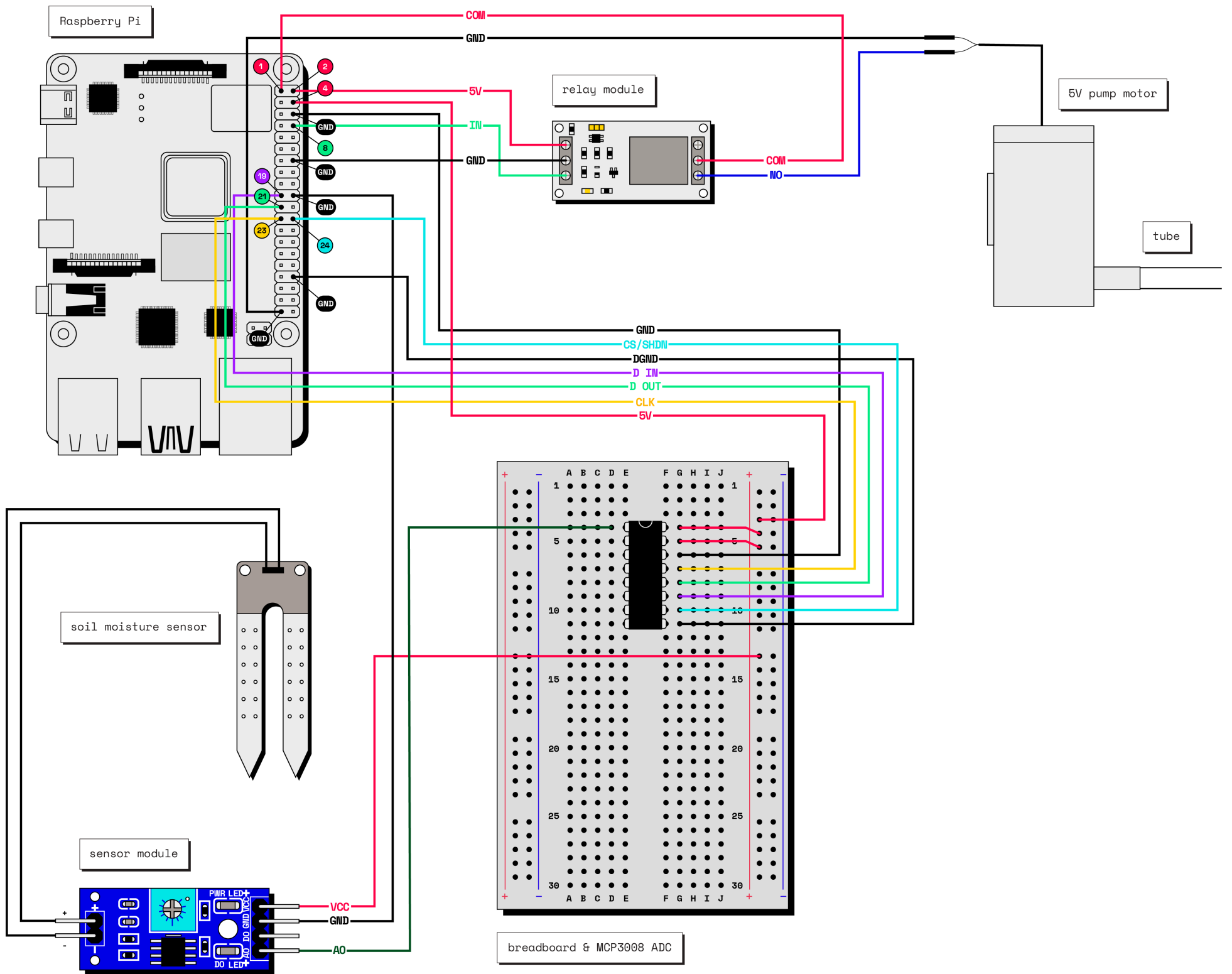
Before starting this tutorial, follow the [Raspberry Pi Setup Guide](#) to prepare your Pi to run `viam-server`. Connect your Pi to its power supply to power it on. Make sure your Pi is flashed with a Viam-compatible operating system, and that you are able to SSH into it.

Set up your plant watering robot

Before programming the Pi to make the plant watering robot functional, you need to physically set up the plant watering robot by wiring the different components together. You will set up the robot to receive signals from the resistive soil moisture sensor and signal to the peristaltic pump when it is time to pump water from the water container to the plant container.

Full wiring diagram

Refer back to this diagram as you complete the steps to wire your hardware. Turn your Pi off while you are wiring the hardware.

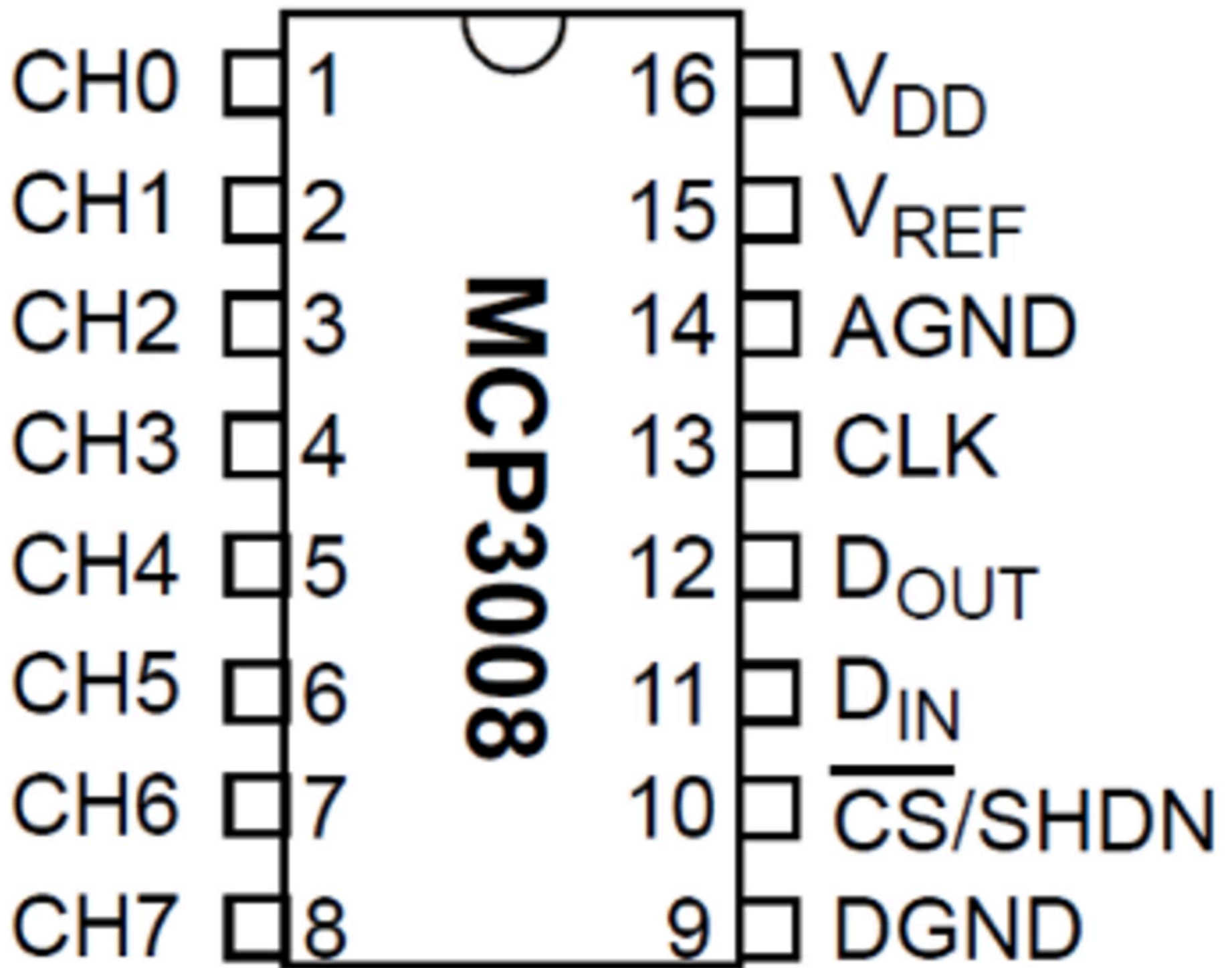


Wire your ADC

The analog-to-digital converter (ADC) converts the resistive soil moisture sensor’s analog readings to digital signals that can be processed by your Pi, which expects digital signals to come to it through its GPIO pins.

Start by wiring your ADC to your Raspberry Pi board.

You can find a Raspberry Pi pinout diagram at pinout.xyz. Reference the following pinout diagram for your MCP3008 analog-to-digital converter:



Tip

The half circle shown in the pinout diagram above should be physically present on your ADC. Use this to orient the ADC to determine the location to insert your wires.

Insert the MCP3008 into your breadboard so that it bridges both sides of the divide. Then, use the rows on the side of your MCP3008’s pins and the GPIO pins on your Pi to connect the pins with wires as follows:

MCP3008 ADC Pin	Raspberry Pi Pin
VDD	5V
VREF	5V
AGND	GND
DGND	GND
CLK	SCLK
DOUT	MISO
DIN	MOSI

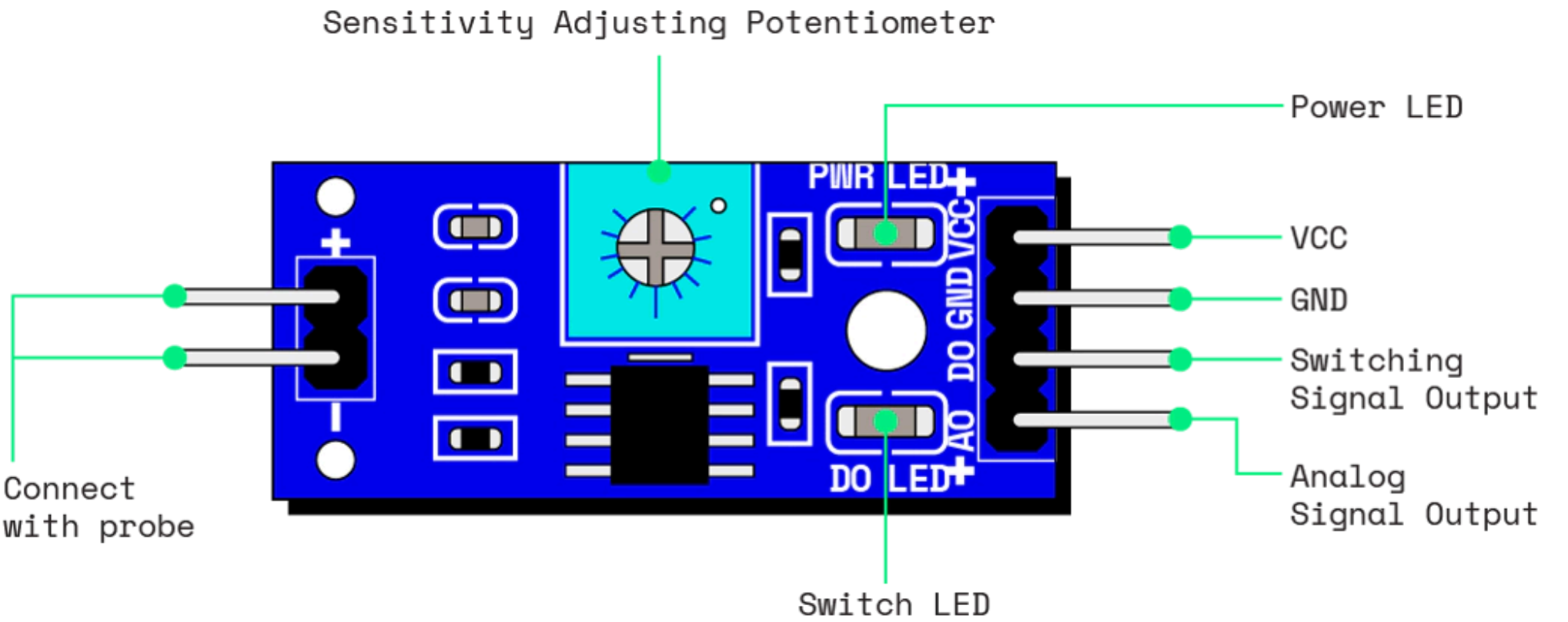
MCP3008 ADC Pin	Raspberry Pi Pin
CS/SHDN	24GPIO8

Use an additional wire to wire Raspberry Pi [pin 4 \(a 5 volt power pin\)](#) to the power rail of the breadboard (the red plus sign column). This brings the 5V power output from the Raspberry Pi to the ADC and the sensor.

Wire your resistive soil moisture sensor

Next, wire your [resistive soil moisture sensor](#) to your Pi and ADC.

Reference this diagram of the blue module part of the sensor:



Start by connecting the female jumper wires at the end of the sensor prongs to the blue module where the diagram shown above is labeled “Connect with Probe.” Be careful of the positive and negative sides, and make sure to match them correctly.

Then, wire the rest of the pins on the module to the Pi and ADC as follows:

Pi		ADC	
Moisture Sensor Pin	Raspberry Pi Pin	Moisture Sensor Pin	MCP3008 ADC Pin
VCC	5V on the power rail	A0 (Analog Signal Output)	CH0
GND	GND		

Put the soil moisture sensor inside of the container holding your plant.

Wire your pump

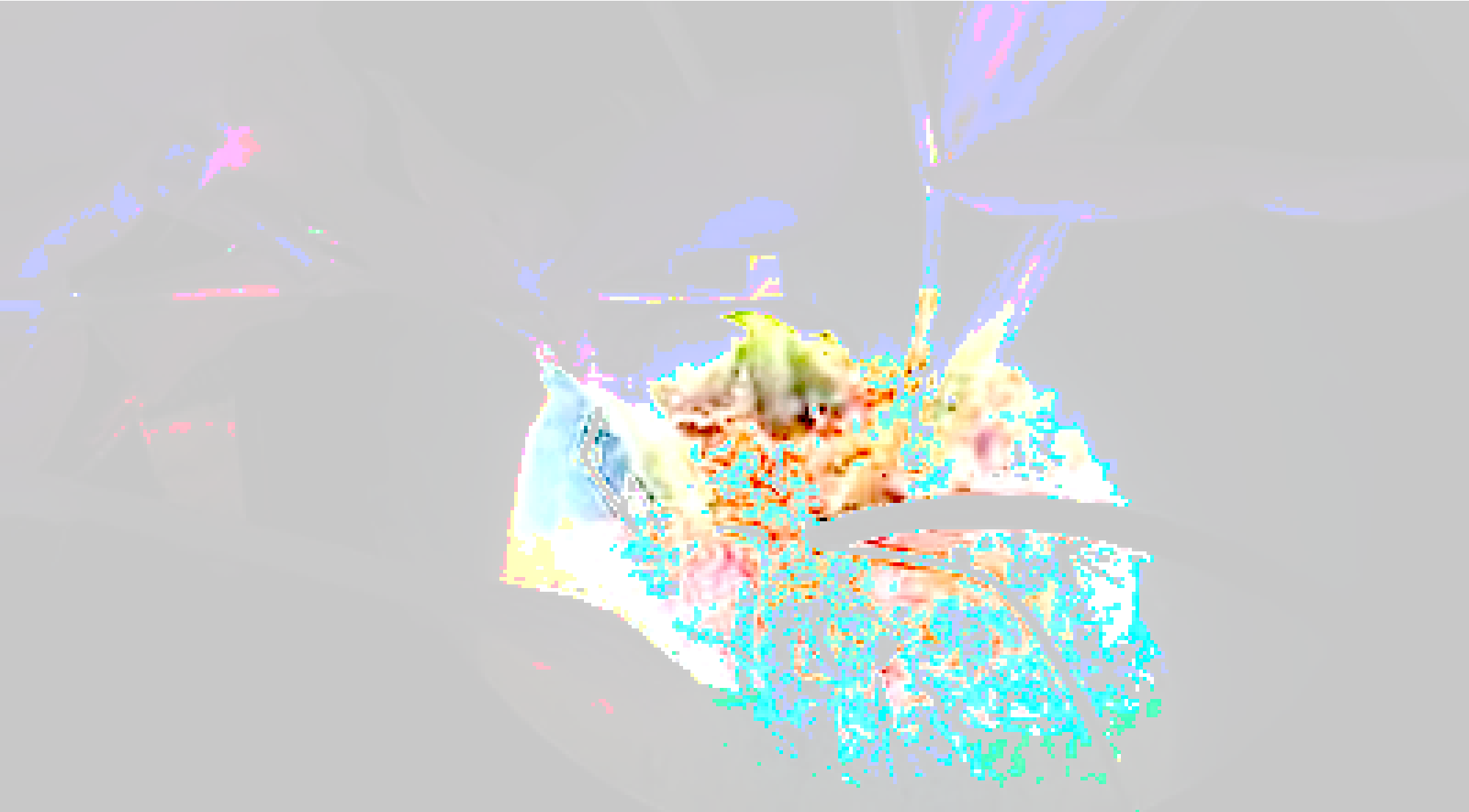
Now, wire and power your pump and relay module to complete your hardware setup:

- 1 Attach an [alligator wire clip](#) to your 5V pump motor’s positive wire, and connect it to the NO pin on relay module. NO stands for normally open, which will keep the circuit open unless the pin is triggered.
- 2 Attach another [alligator wire clip](#) to your 5V pump motor’s negative wire, and connect it to [pin 39 \(ground\)](#) on the Raspberry Pi.
- 3 Connect the COM (common) pin on the relay to [pin 1 \(3.3V\)](#) on the Pi.
- 4 Connect the 5V pin on the relay to [pin 2 \(5V\)](#) on the Pi.
- 5 Connect the GND pin on the relay to [pin 14 \(ground\)](#) on the Pi.
- 6 Connect the IN pin on the relay to the [pin 8 \(GPIO 14\)](#) on the Pi.

Tip

To complete the steps, insert the ends of the jumper wires into the pin gates on the relay module and tighten the screws on these gates with your screwdriver to close the wires inside.

Program your plant watering robot



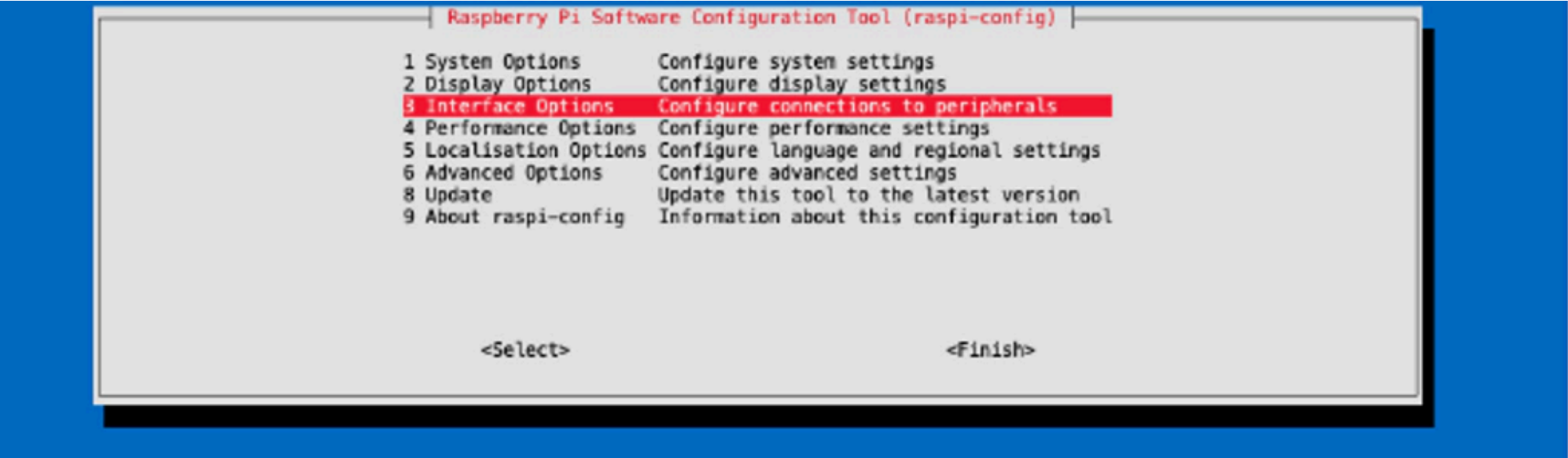
Enable SPI on your Pi

Now that you have wired your ADC and moisture sensor, make sure that the Serial Peripheral Interface (SPI) is enabled on your Pi. Enabling this protocol is necessary to allow the Pi to communicate with the moisture sensor peripheral.

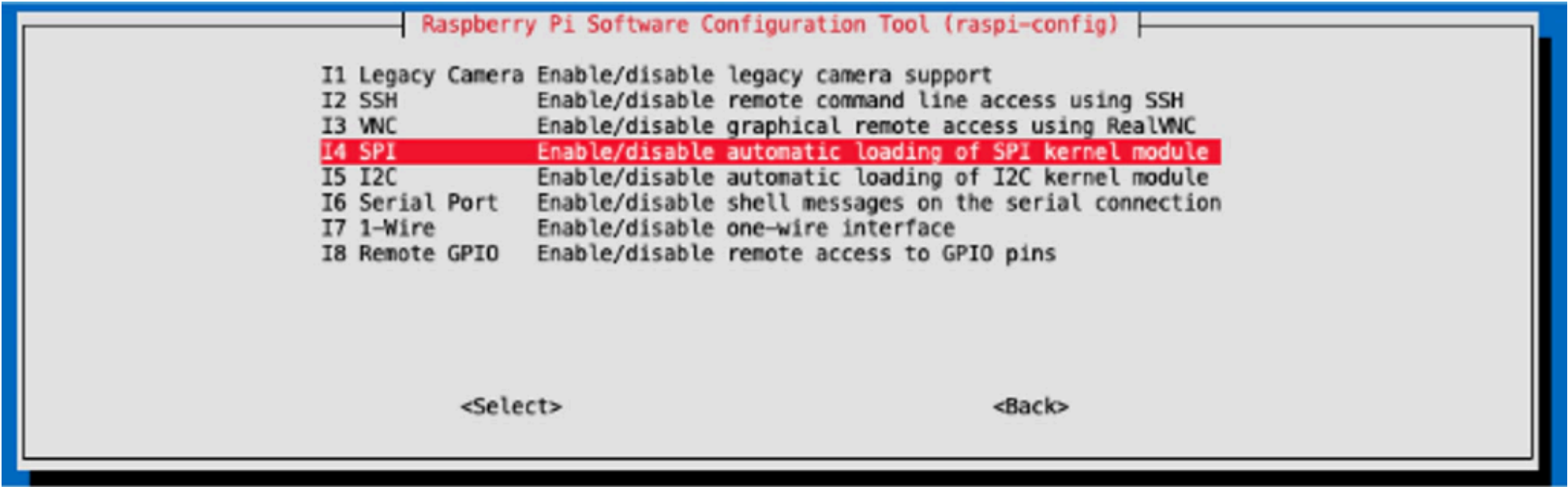
Turn your Pi back on. SSH into your Pi and run the following command:

```
sudo raspi-config
```

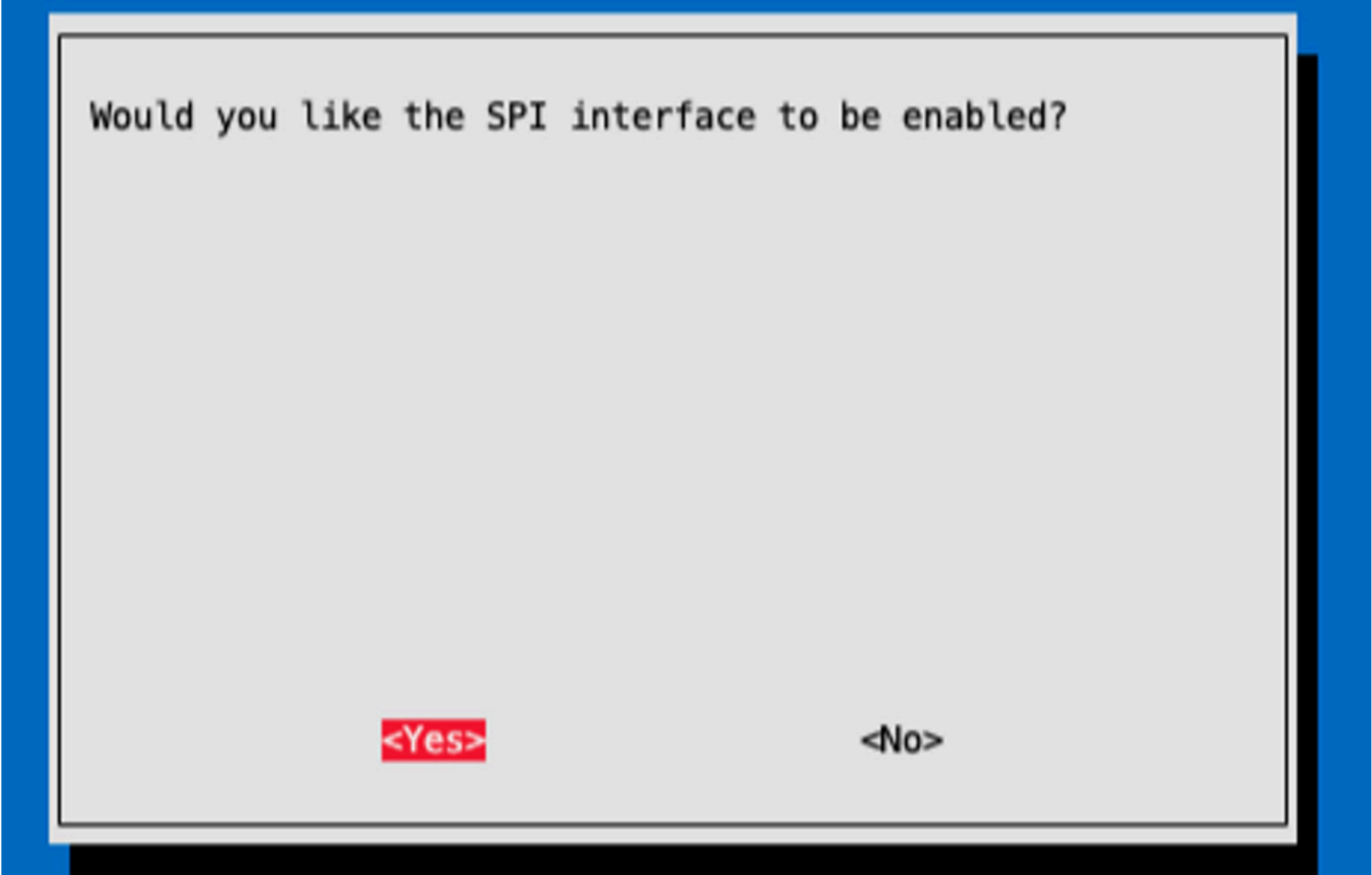
Once the `raspi-config` interface is open, navigate to **Interface Options**:



Then, select **SPI**:



Now, select **Yes** to enable SPI:



Finally, select **Finish**. Restart your Pi using `sudo reboot` to make these changes take effect.

Configure the components of your robot in the Viam app

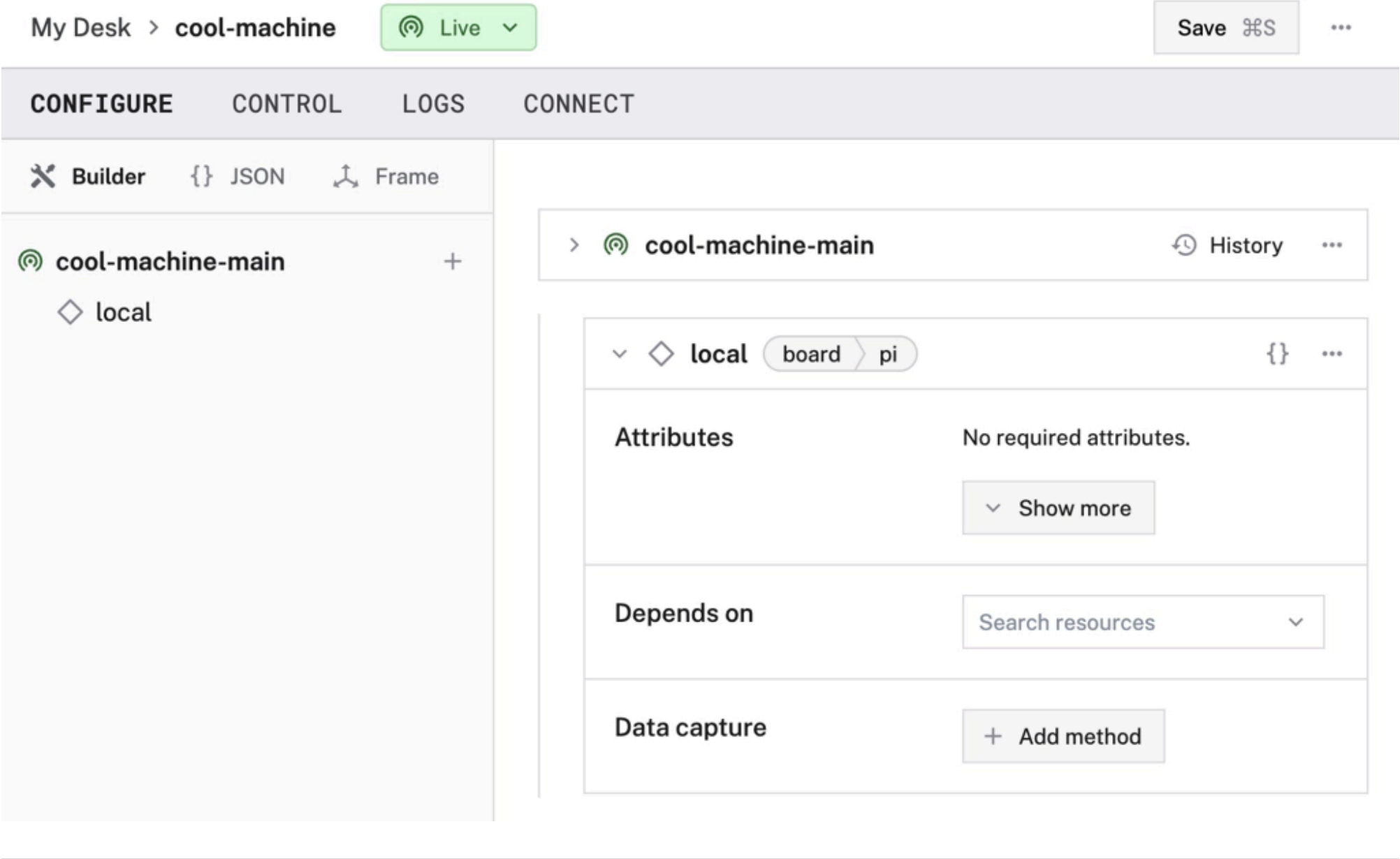
Add a new machine in the [Viam app](#). On the machine’s page, follow the [setup instructions](#) to install `viam-server` on the computer you’re using for your project. Wait until your machine has successfully connected to the Viam app.

Then, navigate to the **CONFIGURE** tab of your new machine’s page in the app.

First, add your Pi as a [board component](#):

[Config Builder](#) [JSON](#)

Click the + icon next to your machine part in the left-hand menu and select **Component**. Select the `board` type, then select the `viam:raspberry-pi:pi` model if you are using a Raspberry Pi 4, Raspberry Pi 3 or Raspberry Pi Zero 2 W. If you are using a Raspberry Pi 5, use the `pi5` model. Enter a name for your board and click **Create**. This tutorial uses the name `local`.



Configure the ADC as a module from the registry

Resources refer to the different **components** and **services** Viam provides for robots to use. *Components* refer to types of hardware, and each component’s built-in **models** support the most common models of this hardware. For example, the [sensor component](#) has an **ultrasonic** model built in for the ubiquitous [ultrasonic sensor](#).

However, there are many different types of sensors used for sensing different things across the [Internet of Things](#). Although the resistive soil moisture sensor is not currently one of Viam’s built-in models, you can add an analog-to-digital-converter (ADC) as a module and use it to get readings from the moisture sensor.

A *module* provides one or more **modular resources**, which add resource types (**components** and **services**) or models that are not built into Viam. A module can be added to your robot from the Viam Registry.

The [Viam Registry](#) allows hardware and software engineers to collaborate on their robotics projects by writing and sharing custom modules with each other. You can add a module from the Viam Registry directly from your robot’s **CONFIGURE** tab in the Viam app, using the **+** (Create) button.

Add the mcp300x-adc-sensor module to your robot in 3 steps:

- 1 Go to your machine’s **CONFIGURE** tab. Click the **+** icon next to your machine part in the left-hand menu and select **Component**.
- 2 Search for **mcp300x** and select **sensor / mcp300x** . Click **Add module**.
- 3 Give your module a name of your choice. We used the name **sensor** .
- 4 Click **Create** to add this module to your machine.
- 5 Find your module’s card on the **CONFIGURE** page. Copy the following JSON object into the configuration field.

```
{
  "channel_map": {
    "moisture": 0
  },
  "sensor_pin": 8
}
```


`sensor_pin` is the GPIO pin number of the Raspberry Pi's pin you connected to the MCP300x chip. If you followed the wiring in this tutorial, this will be `8` which is pin 24, GPIO 8 (SPI Chip Select 0). Otherwise if you are using CS1, use `7`.

Save your config by clicking **Save** in the top right corner of the page.

On the sensor configuration panel, click on the **TEST** area to confirm you can get readings from your sensor.

This module allows you to get multiple readings at the same time from different channels of the ADC sensor. If you wire and configure another sensor, such as a temperature sensor on channel 1, you can add the sensor to the `"channel_map"` and get a reading from it.

Info

If you would like to see how the module works, you can find its code on [GitHub](#).

Now that you have set up your robot, you can put the suction tube of your pump into the water cup, and the output tube into the plant!

Install the Python SDK

Make sure any packages on your Pi are up to date, while connected to your Pi with SSH run:

```
$ sudo apt update
$ sudo apt upgrade
```

Then run the following command to create and activate the virtual environment: If you want to read more on virtual environments, check out [the documentation](#).

```
$ python3 -m venv .venv
$ source .venv/bin/activate
```

Make sure you have `pip` installed for Python 3:

```
$ pip --version
```

If not, run the following command:

```
$ sudo apt install python3-pip
```

Run the following command to install the SDK:

```
$ pip3 install viam-sdk
```

Add Python control code

Follow these instructions to start working on your Python control code:

- 1 Navigate to your machine's page in the [Viam app](#), and click on the **CONNECT** tab and the **Code sample** page.
- 2 Select **Python** as the language.
- 3 Follow the instructions shown under step 1 on that page to install the SDK.
- 4 Then, under step 2 on that page, click the copy icon to copy the generated code sample, which establishes a connection with your robot when run.

To show your machine's API key in the sample code, toggle **Include API key**.

Caution: Keep your API key and machine address safe

We strongly recommend that you add your API key and machine address as an environment variable. Anyone with these secrets can access your machine, and the computer running your machine.

- 5 Paste this code sample into a new file on your Pi.
- 6 Name the file `plant-watering-robot.py`, and save it.

For example, run the following commands on your Pi to create and open the file:

```
source .venv/bin/activate
touch plant-watering-robot.py
nano plant-watering-robot.py
```

Now, you can add code into `plant-watering-robot.py` to write the logic that defines your plant watering system.

To start, add your system logic code into the `main()` function of the program. Use the Viam [board](#) and [sensor](#) API methods to read from the moisture sensor and control the pump's voltage with PWM as a GPIO pin.

You can get your components from the robot like this:

```
# Note that this name, `sensor`, is defined when you add the mcp300x module
sensor = Sensor.from_robot(robot=robot, name='sensor')
# Note that this name, `local`, is defined when you add the board
local = Board.from_robot(machine, "local")
```

And you can add your system logic to run continuously like this:

```
while True:

    # Get the moisture sensor's readings
    readings = await sensor.get_readings()
    soil_moisture = readings.get('moisture')

    # Calculate the average moisture reading from the list of readings
    # to account for outliers
    avg_moisture = sum(soil_moisture) / len(soil_moisture)

    # If the average moisture reading is greater than 60000, trigger pump
    # watering
    if (avg_moisture > 60000):
        print('this plant is too thirsty! giving it more water')

        # Get the GPIO pin with PWM output (pin number 8) the water pump is
        # wired to on the board through the relay's IN wire
        pwm_pin = await local.gpio_pin_by_name(name="8")

        # Run the water pump
        # Set the duty cycle to .8, meaning that this pin will be in the
        # high state, powering the pump motor, for 80% of the duration of the
        # PWM interval period
        await pwm_pin.set_pwm(duty=0.8)

        # Wait for 15 seconds
        print('watering')
        time.sleep(15)

        # Stop the pump by setting the duty cycle to 0%
        await pwm_pin.set_pwm(duty=0.0)

        # Wait 60 seconds so that the water can soak into the soil a bit before
        # trying to water again
        print('waiting a little bit for the water to soak in')
        time.sleep(60)
```

Tip

Make sure to import `time` at the beginning of your version of `plant-watering-robot.py` to be able to use `sleep()` !

Save your `plant-watering-robot.py` program with this logic added in, and then run it on your Pi like this:

```
$ sudo python3 plant-watering-robot.py
```

Test your sensor by putting it in air, water, and different soils to see how the values change and determine your baseline for wet and dry values. Also, consider how often you would like to check the moisture levels of the plant, and how long the plant should be watered.

Next steps

Now that you have created your automatic plant watering system with a resistive soil moisture sensor, you can easily use Viam to automate other aspects of your garden. For example, you can use a [light sensor](#) or a [temperature sensor](#), and get readings from other channels of the MCP3008!

You could set up data capture and [graph your sensor data](#), or [create your own custom Typescript dashboard](#) to display your sensor data.

If you build something based on this please share it in our [Community Discord](#) - we'd love to see it.

